

References

Use **R01**, **R03–R06**, **R09**, **R13**, **R17**, **R20** and **R22** as the short visible slide-6 list; this document is the linked backup. The titles below link to sources already used in the research files. “n.d.” means the linked page does not establish a publication year.

Problem statement and Indian road safety data

- **R01.** Smart India Hackathon / MathWorks, “SIH26037 problem statement,” official PS page, 2026. [Link](#).
- **R02.** Ministry of Education Innovation Cell, “SIH 2026 Guidelines,” competition guide, 2026. [College-hosted copy](#). Its printed September date is stale; use the live PS deadline in R01. [P0 check](#).
- **R03.** Ministry of Road Transport & Highways, *Road Accidents in India 2023*, Government of India report, 2023 data. [Link](#).
- **R04.** Haryana Legislative Assembly, “To solve the problem of Stray Cattle,” Assembly answer, n.d. [Link](#). The answer reports the preceding five years; its exact answer date was not established in P1.
- **R05.** Ministry of Fisheries, Animal Husbandry & Dairying / PIB, “Livestock Census,” Lok Sabha reply, 2022, reporting the 2019 census. [Link](#).

Indian traffic behaviour

- **R06.** Suman, Mondal and Lokesh, “Assessment of pedestrian safety margins at unsignalized mid block crossings in Patna,” *Scientific Reports*, 2026. [Link](#).
- **R07.** Sahu, Parganiha and Pati, “A population estimation study reveals a staggeringly high number of cattle on the streets of urban Raipur in India,” *PLOS ONE*, 2021. [Link](#).
- **R08.** Mohan and Chandra, “Investigating the influence of conflicting flow’s composition on critical gap under heterogeneous traffic conditions,” *International Journal of Transportation Science and Technology*, 2021. [Link](#).
- **R09.** Chennai Traffic Data / Rajput et al., “SPT Chennai traffic trajectories,” dataset project and *Transportation Research Part C* study, 2026. [Project](#), [paper link cited in P4–P5](#). File access and licence remain unverified. [P4–P5](#)

Indian datasets

- **R10.** IIIT Hyderabad, “India Driving Dataset (IDD),” dataset portal, n.d. [Link](#). Licence terms appear at download and were not verified in P4–P5.
- **R11.** IDD-3D authors, “IDD-3D: Indian Driving Dataset for 3D Unstructured Road Scenes,” arXiv preprint, 2022. [Link](#).
- **R12.** Shaik et al., “IDD-AW: A Benchmark for Safe and Robust Segmentation of Drive Scenes in Unstructured Traffic and Adverse Weather,” *WACV*, 2024. [Link](#).
- **R13.** Parikh, Saluja, Jawahar and Sarvadevabhatla, “IDD-X: A Multi-View Dataset for Ego-relative Important Object Localization and Explanation in Dense and Unstructured Traffic,” research paper, 2024. [Link](#). Its labels are **not metric future trajectories**.
- **R14.** Bokkasam et al., “Pedestrian Intention and Trajectory Prediction in Unstructured Traffic Using IDD-PeD,” arXiv preprint, 2025. [Link](#). Dataset licence unverified.
- **R15.** Kumar, Reddy and Rajalakshmi, “DriveIndia: An Object Detection Dataset for Diverse Indian Traffic Scenes,” *ITSC*, 2025. [Link](#). TIHAN access requires an agreement; exact reuse terms need checking.
- **R16.** Khoba et al., “A Fine-Grained Vehicle Detection Dataset for Unconstrained Roads,” research paper, 2022. [Link](#).
- **R17.** TIHAN IIT Hyderabad, “TIAND: multimodal Indian driving scenes,” dataset portal and IEEE paper, 2024. [Portal](#), [paper](#). Request and usage agreement required.
- **R18.** Chandra et al., “METEOR: A Dense, Heterogeneous, and Unstructured Traffic Dataset With Rare Behaviors,” arXiv preprint, 2021/2022. [Link](#). Dataset reuse terms need checking.
- **R19.** RoadText authors, “RoadText-1K: Text Detection & Recognition Dataset for Driving Videos,” arXiv preprint, 2020. [Link](#). Dataset licence unverified.

Unfilled data needs: Indian police hand gestures, siren/horn audio with direction labels, and driver-monitoring footage. The design must collect or simulate these and mark synthetic results as such. [P15](#)

Planning and safety prior art; competitor evidence

- **R20.** Kruse, Yel, Senanayake and Kochenderfer, “Uncertainty-Aware Online Merge Planning with Learned Driver Behavior,” arXiv preprint, 2022. [Link](#).

- **R21.** Cai and Hsu, “Closing the Planning-Learning Loop with Application to Autonomous Driving” (*LeTS-Drive*), *IEEE Transactions on Robotics*, 2022. [Link](#).
- **R22.** Le Large et al., “Mosaic: An Extensible Framework for Composing Rule-Based and Learned Motion Planners,” *IROS*, 2026. [Link](#). Separate proposal and verification are prior art.
- **R23.** Trautman and Krause, “Unfreezing the Robot: Navigation in Dense, Interacting Crowds,” research paper, 2010. [Link](#).
- **R24.** “Multi-Agent Reachability Calibration with Conformal Prediction,” arXiv preprint, 2023. [Link](#). Calibration is conditional on its assumptions; it does not rescue missed detections.
- **R25.** ARAI, “Intelligent Vehicle Technology: Indian driving scenario repository for ADAS functions,” institutional page, n.d. [Link](#).
- **R26.** Warwick WMG, “Crowdsourcing Indian Traffic Scenarios for ADAS Development and Testing,” research paper, 2026. [Link](#).
- **R27.** Swaayatt Robots, “Technology and autonomous-driving work,” company site, n.d. [Link](#). Treat detailed performance as company claims.
- **R28.** Minus Zero, “How did a foundational model learn to navigate the busy streets of Bengaluru?,” company technical account, 2025. [Link](#). Its detailed runtime safety mechanism is unverified.

MathWorks and simulation tools

- **R29.** MathWorks, “Client, Reader and Writer blocks for Eclipse SUMO co-simulation,” R2026a documentation. [Client](#), [Reader](#), [Writer](#).
- **R30.** MathWorks, “Set Up and Connect to CARLA Simulator,” product documentation, n.d. [Link](#).
- **R31.** CARLA project, “Windows build; add a new vehicle; add new props,” CARLA 0.9.15 documentation, n.d. [Build](#), [vehicle](#), [props](#).
- **R32.** MathWorks, “Verify MEX Functions in the MATLAB Coder App,” product documentation, n.d. [Link](#).
- **R33.** MathWorks, “Passenger Vehicle Dynamics Models,” Vehicle Dynamics Blockset documentation, n.d. [Link](#).